~~$\delta_K(a) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{ika}\,dk = \frac{1}{2\pi}\left(\frac{1}{ia}e^{iaK}\right)\Big|_{K=-\infty}^{K=+\infty}$~~

$$\delta_K(a) = \frac{1}{2\pi}\int_{-K}^{+K} e^{ika}\,dk = \frac{1}{2\pi}\left[\frac{1}{ia}e^{ika}\right]\Big|_{k=-K}^{k=+K}$$

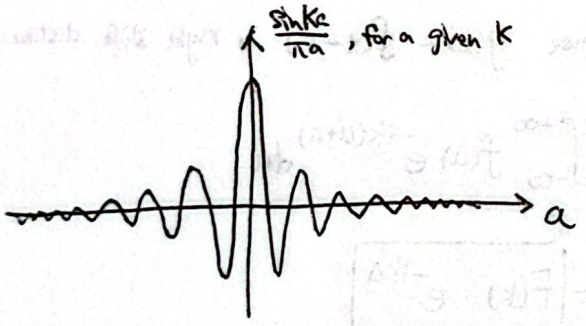
$$= \frac{1}{2\pi ia}\left(e^{iKa} - e^{-iKa}\right)$$

$\cos Ka + i\sin Ka - (\cos Ka - i\sin Ka) = 2i\sin Ka$

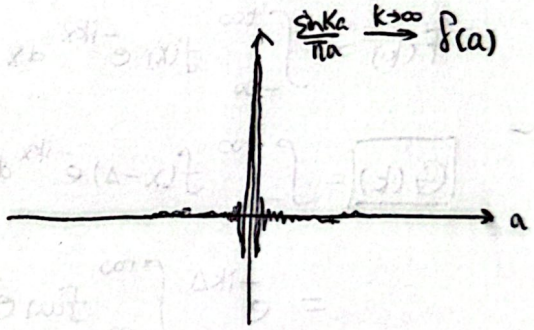
$$= \frac{\sin Ka}{\pi a}$$

~~$\int_{K\to\infty} \delta_K(a)\,da = \int_{K\to\infty} \frac{\sin Ka}{\pi a}$~~ , for a given ~~Large~~ $K$ , $\delta_K(0) = \left.\frac{\sin Ka}{\pi a}\right|_{a=0} = \left.\frac{K\cos Ka}{\pi}\right|_{a=0} = \frac{K}{\pi}$

$\therefore \lim_{K\to\infty} \delta_K(0) = \lim_{K\to\infty} \frac{K}{\pi} = \infty$ (or $\frac{\sin Ka}{\pi a} \overset{a\to 0}{=} \frac{Ka}{\pi a} = \frac{K}{\pi}$)




$K\to\infty$ ⇒




Besides. for $\forall K>0$, $\int_{-\infty}^{+\infty} \delta_K(a)\,da = 1$

Proof: $\int_{-\infty}^{+\infty} \frac{\sin Ka}{\pi a}\,da = \int_{-\infty}^{+\infty} \frac{\sin Ka}{\pi(Ka)}\,d(Ka) = \int_{-\infty}^{+\infty} \frac{\sin u}{\pi u}\,du$ → Dirichlet Integral.

Introduce $F(s) = \int_{-\infty}^{+\infty} e^{-su}\frac{\sin u}{\pi u}\,du$ ; $\int_{-\infty}^{+\infty} \frac{\sin Ka}{\pi a}\,da = \lim_{s\to 0} F(s)$

$$F'(s) = \frac{1}{\pi}\int_{-\infty}^{+\infty} -e^{-su}\sin u\,du = -\frac{1}{\pi}\frac{2}{s^2+1} \Rightarrow F(s) = \left(-2\arctan(s) + C\right)\frac{1}{\pi}$$

(this is $2\mathcal{L}(\sin u)$)

Since $\lim_{s\to\infty} F(s) = 0 = -2\cdot\arctan(\infty) + C = -\pi + C \Rightarrow C = \pi$ ; $F(s) = \left(-2\arctan(s) + \pi\right)\frac{1}{\pi}$

$\therefore \int_{-\infty}^{+\infty} \delta_K(a)\,da = \int_{-\infty}^{+\infty} \frac{\sin u}{\pi u}\,du = F(0) = 1$ . □ therefore, $\delta_K(a) \xrightarrow{K\to\infty} \delta(a)$

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Fourier Side: why do we need a "$\frac{1}{2\pi}$" when doing reverse transform?

trans: $\int_{-\infty}^{+\infty} f(y)\, e^{-iky}\, dy = F(k)$

inverse-trans: $\frac{1}{2\pi}\int_{-\infty}^{+\infty}\left[\int_{-\infty}^{+\infty} f(y)\, e^{-iky}\, dy\right] e^{ikx}\, dk = f(x)$

$$= \boxed{\frac{1}{2\pi}} \int_{-\infty}^{+\infty} f(y)\left[\int_{-\infty}^{+\infty} e^{ik(x-y)}\, dk\right] dy$$

$$= \int_{-\infty}^{+\infty} f(y)\, \delta(x-y)\, dy = f(x)$$

"$\frac{1}{2\pi}$" is for constructing the delta function.

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How Fourier transform describes image shifting? take 1D signal for example:

$F(k) = \int_{-\infty}^{+\infty} f(x)\, e^{-ikx}\, dx$ , suppose $g(x) = f(x-\Delta)$ , right shift distance $= \Delta$

$$\boxed{G(k)} = \int_{-\infty}^{+\infty} f(x-\Delta)\, e^{-ikx}\, dx \overset{u=x-\Delta}{=\!=} \int_{-\infty}^{+\infty} f(u)\, e^{-ik(u+\Delta)}\, du$$

$$= e^{-ik\Delta}\int_{-\infty}^{+\infty} f(u)\, e^{-iku}\, du = \boxed{F(k)\cdot e^{-ik\Delta}}$$

if $F(k) = A(k)\, e^{i\phi(k)}$ (complex form), then $G(k) = A(k)\, e^{i[\phi(k)-k\Delta]}$ . just change the angle.

Also note that: $\boxed{g(x) = f(x-\Delta) = f(x) * \delta(x-\Delta)}$ ↙ key.

proof: $f(x) * \underbrace{\delta(x-\Delta)}_{h(x)} = \int_{-\infty}^{+\infty} f(u)\, h(x-u)\, du = \int_{-\infty}^{+\infty} f(u)\, \delta(x-u-\Delta)\, du = f(x-\Delta)$. □

Therefore, $\underbrace{G(k)}_{\text{known}} = F(k)\,\mathcal{F}[\delta(x-\Delta)] = F(k)\int_{-\infty}^{+\infty}\delta(x-\Delta)\, e^{-ikx}\, dx = \underbrace{F(k)}_{\text{known}}\, e^{-ik\Delta}$

How to calculate $\Delta$? **Normalized Cross-Power Spectrum**

$$G(k) = F(k)\, e^{-ik\Delta}, \quad F(k) = A(k)\, e^{i\phi(k)}$$

$$R(k) := \frac{G(k)F^*(k)}{|G(k)F^*(k)|} = \frac{A(k)\, e^{i[\phi(k)-k\Delta]}\, A(k)\, e^{-i\phi(k)}}{A^2(k)} = e^{-ik\Delta}$$

$$\mathcal{F}^{-1}[R(k)] = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{-ik\Delta}\cdot e^{iks}\, dk = \frac{1}{2\pi}\int_{-\infty}^{+\infty} e^{ik(s-\Delta)}\, dk = \delta(s-\Delta)$$

$$\therefore \boxed{\Delta = \underset{s}{\operatorname{argmax}}\, \mathcal{F}^{-1}[R(k)]}$$ this argmax is easy to calculate due to the sharp $\delta$ function (better than simple cross-correlation)

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